

AI Model Comparison for Paper Extraction

Claude vs GPT vs Gemini: cost and quality for reading past papers (questions, LaTeX maths, tables, diagrams, mark schemes)

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Summary. For best quality, the three top models are **Claude Opus 5.5**, **GPT-5.5** and **Gemini 3.1 Pro**. On our real papers, one paper costs about **\$1.05–1.45 (Claude Opus 5.5)**, **\$1.35–1.95 (GPT-5.5)** and **\$0.55–0.75 (Gemini 3.1 Pro)**. The mid tier (Claude Sonnet 5.5, GPT-5.4, Gemini 3.5 Flash) costs about half of that. **Diagrams add almost nothing to the AI cost:** the cost depends on the number of pages, not on what is drawn on them. Claude is the only model already **proven on our papers** (the 10 papers in the app were extracted with Claude). The others need a side-by-side test on 2–3 papers before we choose. Even the most expensive option is under \$2 per paper, so **quality should decide, not price**.

~30

question-paper pages per paper (24–36), ~21 with questions

~19

mark-scheme pages per paper (12–22)

~16

diagrams per paper (9–21)

~31 / 17

question parts / whole questions per paper

Measured from the 10 Edexcel AS Physics (8PH0) papers already in the system (2015–2024).

1. How the cost is calculated

AI services charge per **token** (about ¾ of a word). Reading (input) is cheap; writing (output) costs more. A page picture is counted as a fixed number of tokens based on its **size**, not its content.

What our upload sends for one paper

Part	How it is sent	AI requests
Question-paper pages	1 page per request (blank / cover pages skipped), high image detail	~28
Mark-scheme pages	2 pages per request	~10
Instructions sent with every request	extraction rules + the subject's topic list	~2,000 tokens each

Tokens per page picture (our pages are 1190 × 1683 pixels)

Provider	Approx. tokens per page	Note
Claude	~1,550	images are scaled to a maximum size, then counted by area
GPT (high detail)	~1,100	counted in tiles; high detail is needed for small labels and subscripts
Gemini (high resolution)	~1,120	medium resolution would be ~560, but small text may be misread

Total tokens for one typical paper

Provider	Input (reading)	Output: questions + answers	Output: model "thinking"
Claude	~149,000	~13,000	~10,000–30,000
GPT	~128,000	~13,000	~10,000–30,000
Gemini	~129,000	~13,000	~10,000–30,000

Output size comes from our real data (~8,500 characters of questions + ~8,200 characters of mark scheme per paper, plus the structured format and diagram positions). Top models "think" before answering and that thinking is billed as output; the range depends on the effort setting. Prices below include it as a range.

2. Diagrams: how much they cost

AI cost of diagrams: almost zero

- A page costs the same whether it has a diagram or not; the price is set by the picture size.
- The AI only returns a small "box" (position) for each diagram: ~40 tokens each, ~650 tokens per paper (under \$0.02 even on the most expensive model).
- Cutting the diagram out of the page is done **on our own server**, free.

Where diagrams do matter: quality

- Diagrams have small labels, units and subscripts, so pages must be sent at **high detail**. Lower detail saves 40–90% of reading cost but risks mistakes.
- The AI must place the diagram box correctly; a wrong box means an admin redraws it (the review screen allows this).
- Tables that are diagrams are cropped as pictures, not typed twice (rule already added).

Conclusion: cost grows with **pages**; quality depends on **diagrams, maths and tables**. We keep high detail.

3. Price list (per 1 million tokens)

Tier	Model	Input	Output	Use
Top	Claude Opus 5.5	\$4.00	\$20.00	best quality (Anthropic official price)
Top	GPT-5.5	\$5.00	\$30.00	best GPT quality
Top	Gemini 3.1 Pro	\$2.00	\$12.00	best Gemini quality
Mid	Claude Sonnet 5.5	\$2.00	\$10.00	strong, half of Opus
Mid	GPT-5.4	\$2.50	\$15.00	strong, half of GPT-5.5
Mid	Gemini 3.5 Flash	\$1.50	\$9.00	fast, good value
Low	Claude Haiku 4.5	\$1.00	\$5.00	cheap, weaker on hard pages
Low	GPT-5.4 mini	\$0.75	\$4.50	cheap
Low	GPT-5 mini	\$0.25	\$2.00	very cheap, weakest
Low	Gemini 3.5 Flash-Lite	\$0.30	\$2.50	very cheap, weakest

Claude prices: Anthropic official. GPT and Gemini prices: recent public pricing pages (see sources). Not included: GPT-5.4 pro (\$30 / \$180) and Claude Fable 5.1 (\$10 / \$50), which cost far more than this job needs.

4. Cost per paper (typical paper: ~30 QP pages + ~19 MS pages)

Tier	Model	Per paper	100 papers	1,000 papers	10,000 papers
Top	Claude Opus 5.5	\$1.05–1.45	\$105–145	\$1,050–1,450	\$10.5k–14.5k
Top	GPT-5.5	\$1.35–1.95	\$135–195	\$1,350–1,950	\$13.5k–19.5k
Top	Gemini 3.1 Pro	\$0.55–0.75	\$55–75	\$550–750	\$5.5k–7.5k
Mid	Claude Sonnet 5.5	\$0.55–0.75	\$55–75	\$550–750	\$5.5k–7.5k
Mid	GPT-5.4	\$0.65–0.95	\$65–95	\$650–950	\$6.5k–9.5k
Mid	Gemini 3.5 Flash	\$0.40–0.60	\$40–60	\$400–600	\$4k–6k
Low	Claude Haiku 4.5	~\$0.20–0.25	\$20–25	\$200–250	\$2k–2.5k
Low	GPT-5.4 mini	\$0.20–0.30	\$20–30	\$200–300	\$2k–3k
Low	GPT-5 mini	\$0.08–0.12	\$8–12	\$80–120	\$800–1.2k
Low	Gemini 3.5 Flash-Lite	\$0.10–0.15	\$10–15	\$100–150	\$1k–1.5k

Save ~50% with batch mode. All three providers offer a batch option (results within hours instead of seconds) at about half price. Admin uploads do not need instant results, so batch would bring Claude

Opus 5.5 to about **\$0.55–0.75 per paper**. Prompt caching (the same instructions on every page) can also reduce the reading cost.

Paper Analyzer · AI model comparison for paper extraction · 8 October 2026 · Prices are list prices and change often; check official pages before purchase.

Effect of paper size (top-tier models, mid thinking level)

Paper size	Claude Opus 5.5	GPT-5.5	Gemini 3.1 Pro
Small: 20 QP pages + 10 MS pages	~\$0.80	~\$1.05	~\$0.42
Typical: 30 QP pages + 19 MS pages (our papers)	~\$1.25	~\$1.65	~\$0.65
Large: 40 QP pages + 30 MS pages	~\$1.85	~\$2.45	~\$1.00

Number of diagrams does not change these figures (see section 2). A paper with more pages costs more because every page is read.

5. Quality comparison

Proven = tested on our own papers. **Expected** = based on the model's general strength; must be confirmed by our test.

What we need	Claude Opus 5.5	GPT-5.5	Gemini 3.1 Pro
Maths written correctly as LaTeX (units, powers, fractions)	Proven: excellent	Expected: very good	Expected: very good
Multi-part questions (a), (b)(i)... kept in order	Proven: excellent	Expected: very good	Expected: good
Tables (rows, columns, empty working columns)	Proven: very good	Expected: very good	Expected: good
Finding diagram positions (crop boxes)	Proven: good	Expected: good	Expected: very good
Matching mark schemes to the right question part	Proven: excellent	Expected: very good	Expected: good
Following the exact output format every time	Proven: excellent	Expected: excellent	Expected: very good
Topic / subtopic matching to the syllabus	Proven: very good	Expected: very good	Expected: very good
Speed per paper	moderate	moderate	fast
Cost per paper	\$1.05–1.45	\$1.35–1.95 (highest)	\$0.55–0.75 (lowest)

Mid and low tiers: what to expect

Tier	Models	Expected quality for our papers	Advice
Mid	Claude Sonnet 5.5, GPT-5.4, Gemini 3.5 Flash	Close to the top tier on most pages; slightly more mistakes on dense maths, long tables and tricky mark-scheme matching.	Test first may be the best value
Low	Claude Haiku 4.5, GPT-5.4 mini, GPT-5 mini, Gemini 3.5 Flash-Lite	Noticeably more errors in LaTeX, subscripts, table cells and diagram boxes; more admin fixing.	Not recommended for extraction

Hidden cost: admin time. If a cheaper model saves \$0.50 per paper but an admin spends 20 extra minutes fixing it, the cheaper model is far more expensive overall. That is why quality should decide.

6. Current setup in the app

Item	Now
10 papers in the app	Extracted with Claude by a developer (Claude Code, no API cost), then imported
Admin upload page	Uses Gemini (gemini-flash-latest, backups 3.5 Flash and Flash-Lite) on a free key (5 requests per minute; slow for real use)
API keys	Gemini: yes (free tier) · OpenAI: none · Claude: none
Similar questions (embeddings)	Free model on our own server; no key and no cost; not affected by this choice

7. Recommendation

1. **Run a fair test:** upload the same 2–3 papers (one with many diagrams, one with tables) through **Claude Sonnet 5.5, GPT-5.4 and Gemini 3.5 Flash**, and compare against our existing Claude versions (maths, tables, parts, diagram boxes, mark schemes).
2. If a mid-tier model is close to Claude's quality, use it (about **\$0.40–0.95 per paper**). If not, use a top model: **Claude Opus 5.5** (proven) or **Gemini 3.1 Pro** (lowest top-tier price).
3. Use **batch mode** to halve the cost and a **paid key** so uploads are not slowed by free-tier limits.
4. Adding Claude to the upload page needs an **Anthropic API key** and a small developer change (the upload already has a provider switch). After that, admins upload with no developer involvement.
5. Keep pages at **high detail** for diagram-heavy subjects; never use the low tier for extraction.

8. Assumptions and sources

- Paper statistics are from our 10 Edexcel AS Physics papers. Other subjects and boards may have more or fewer pages; cost scales with pages.
- Image token counts are approximate and depend on each provider's resizing rules and detail settings.
- "Thinking" output varies with the effort setting; shown as a range (10,000–30,000 tokens per paper).
- Claude prices are Anthropic's official API prices. GPT and Gemini prices are from public pricing pages (October 2026) and may change; confirm on the official OpenAI and Google pricing pages before purchase.
- Sources: morphllm.com/openai-api-pricing · intuitionlabs.ai (ChatGPT API pricing 2026) · morphllm.com/gemini-api-pricing · anotherwrapper.com (GPT-5 mini, GPT-5.4 mini, Gemini 3.5 Flash, Flash-Lite pricing) · cloudzero.com/blog/gemini-pricing.